Lab 1: Git Installation & Setup

1: git --version

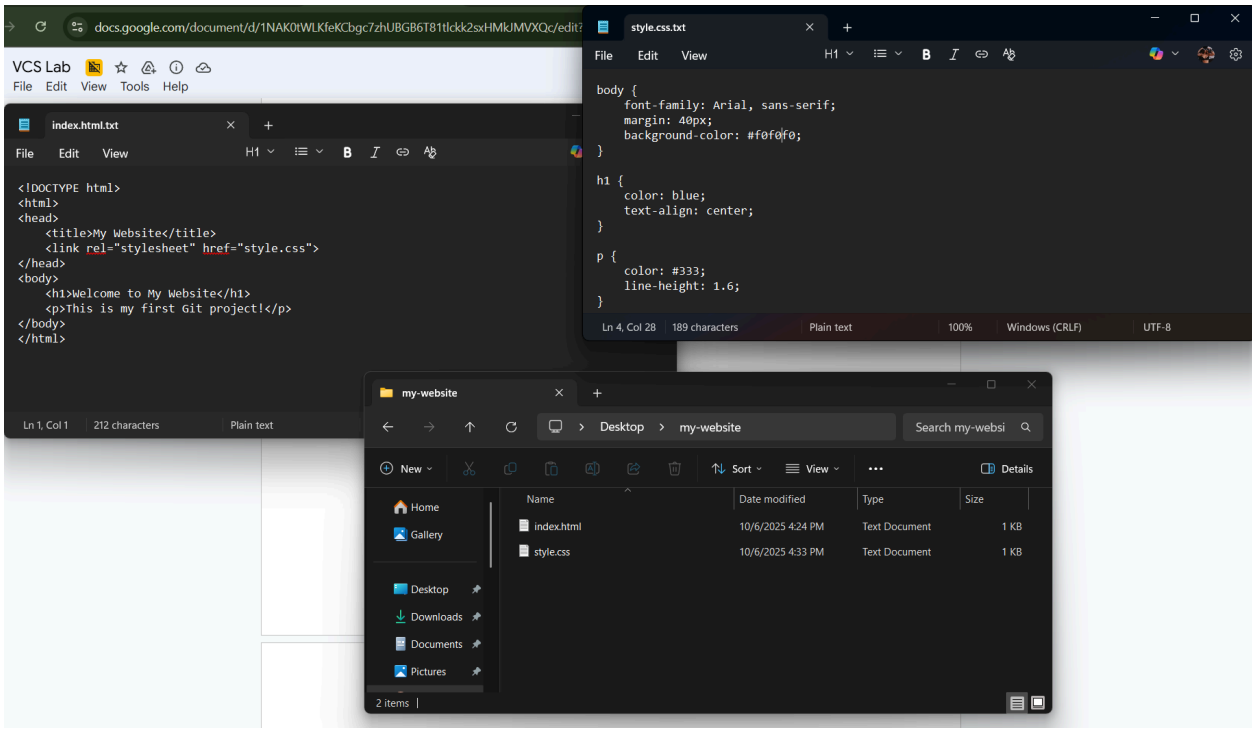
3: git config --global user.name "Riwaj Dangol"
  git config --global user.email "riwaj.dangol@patancollege.edu.np"

4: Setting the username and gmail is important in Git so that each commit made can be identified.

5:

```
C:\WINDOWS\system32\cmd.

C:\Users\ACER\Desktop\my-website>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Riwaj Dangol
user.email=riwaj.dangol@patancollege.edu.np
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/Riwaj0412/my-website.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.master.remote=origin
branch.master.merge=refs/heads/master
branch.improve-styling.remote=origin
branch.improve-styling.merge=refs/heads/improve-styling
```

Lab 2: Create a Local Website Project



Lab 3: Initialize Git Repository

1: “cd” command is used to navigate to the folder

2: git init

3: .git folder is a hidden folder which tracks all the changes made.

4:

```
C:\Users\ACER\Desktop\my-website>git status
On branch master
```

5:

```
C:\Users\ACER\Desktop\my-website>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Riwaj Dangol
user.email=riwaj.dangol@patancollege.edu.np

C:\Users\ACER\Desktop\my-website>git init
Initialized empty Git repository in C:/Users/ACER/Desktop/my-website/.git/

C:\Users\ACER\Desktop\my-website>git status
On branch master

No commits yet
```

Lab 4: Add and Commit Files

1: The purpose of the staging area in git is to review and prepare the changes before saving them permanently to the repository.

2: git add index.html.txt

3: git commit -m "This is a change in HTML"

4: git add style.css.txt

git commit -m "This is a change in CSS"

5: The commit ID and the message shown is mentioned below:

```
commit b50016232388a74b9bf9680fa5a792ef418d0657 (HEAD -> master)
Author: Riwaj Dangol <riwaj.dangol@patancollege.edu.np>
Date:   Mon Oct 6 16:25:42 2025 +0545
```

```
    Add CSS styling for website

commit fcd48966c3fed5b6a307035d91e52fa20934af41
Author: Riwaj Dangol <riwaj.dangol@patancollege.edu.np>
Date:   Mon Oct 6 16:25:06 2025 +0545

    Add initial HTML file for website
```

6 :

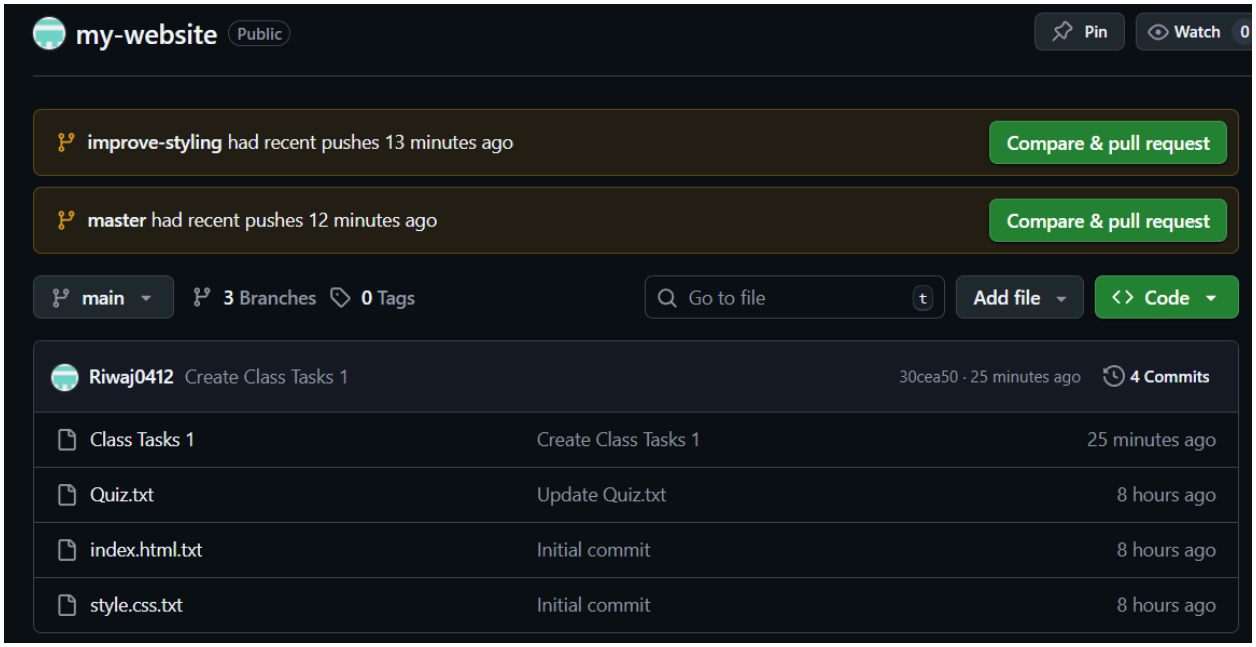
```
C:\Users\ACER\Desktop\my-website>git log
commit b50016232388a74b9bf9680fa5a792ef418d0657 (HEAD -> master)
Author: Riwaj Dangol <riwaj.dangol@patancollege.edu.np>
Date:   Mon Oct 6 16:25:42 2025 +0545

    Add CSS styling for website

commit fcd48966c3fed5b6a307035d91e52fa20934af41
Author: Riwaj Dangol <riwaj.dangol@patancollege.edu.np>
Date:   Mon Oct 6 16:25:06 2025 +0545

    Add initial HTML file for website
```

Lab 5: Create GitHub Repository and Push Code

3: git remote add origin https://github.com/Riwaj0412/my-website.git

5: It helps to upload the commits made to a remote repository and also helps to set up the tracking between the local and remote branches

6:

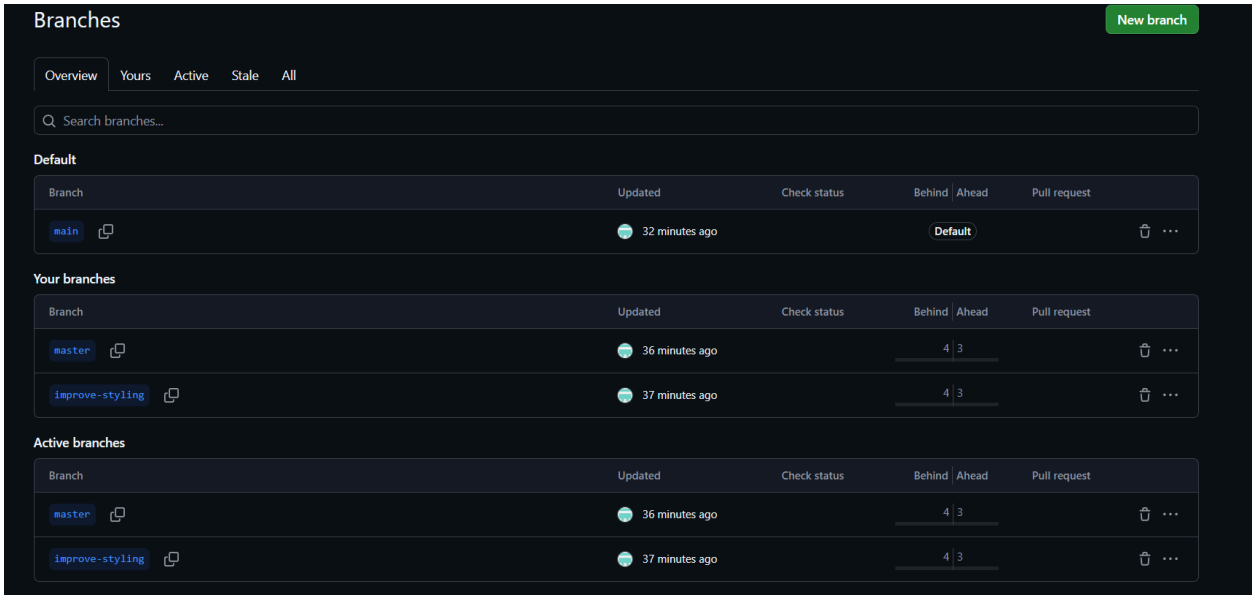


Lab 6: Create and Switch Branches

1: Branch is like a separate timeline where it lets one work on a different version of the project independently without affecting the main code.

3: git branch

7:



Lab 7: Merge Branches

1: Merging basically means combining one branch into another where a new work is usually brought into the main line of development.

2: git checkout master

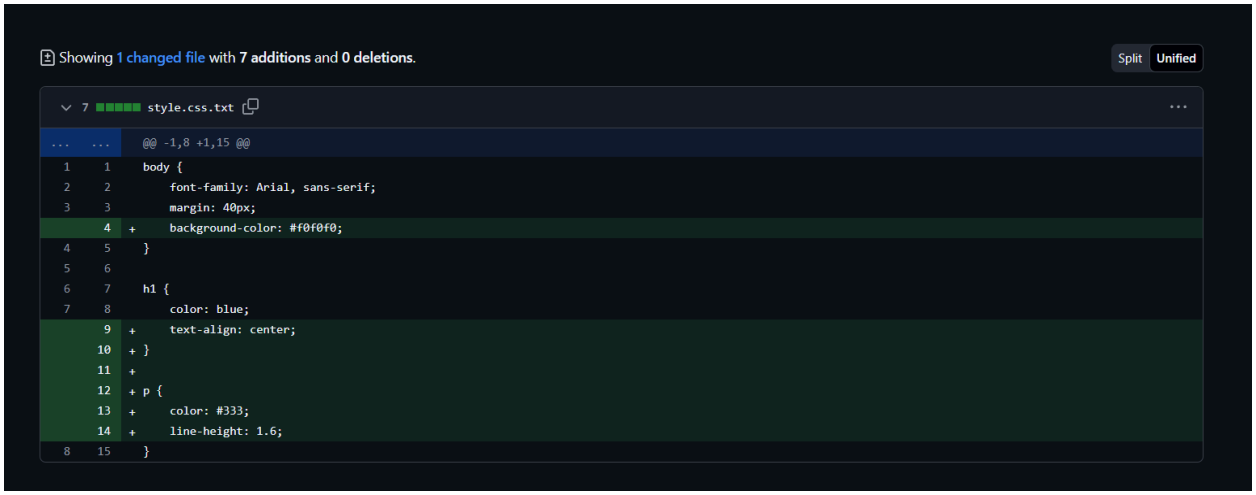
5: After merging, the feature branch's changes are integrated into the main branch, the branch itself remains but can safely be deleted — the work it contained is not lost.

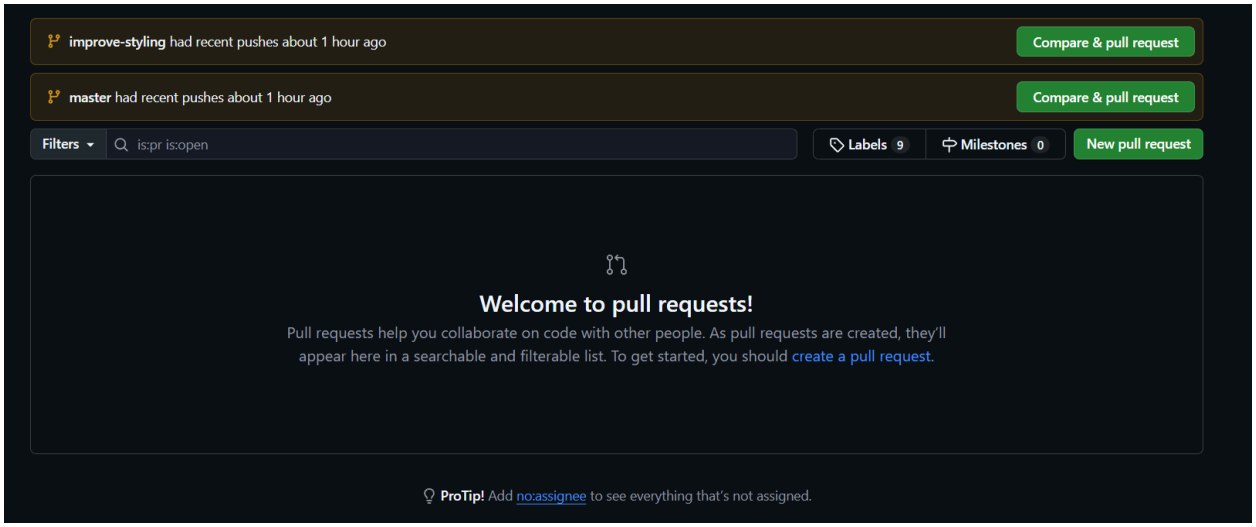
6:

Lab 8: Collaboration Simulation

1: To fork basically means to creating a copy of someone else's repository in our own GitHub account

7:





Lab 9: Reflection Task

1: Git is basically a version control system where the change in code can be tracked, collaborate with others or to manage the project history.

2: git add basically is preparing for the changes in the next commit,
Git commit is used to record the changes in the local Git repository
Git push is used to send the local commits to a remote repository like GitHub

3: Branch is like a separate timeline where it lets one work on a different version of the project independently without affecting the main code.
Developers use branches so that they can try out new things safely with efficient collaboration with others while keeping the code stable.

4: git log

5: Doing the specific merge and pushing it to the remote repository.

6: The skill to create a new features in the code without hampering the main code is what I have I learned today which will help me in the future.